TABLEAU ASSIGNMENT

Calculated Fields, Calculation Levels, and Null Values

1) What is a Calculated Field in Tableau? How is it different from a regular field?

A **Calculated Field** in Tableau is a new field created by writing a formula that uses existing data fields, constants, and functions. It helps users create new measures or categories without changing the original source data. For example, if a dataset contains Sales and Cost, a calculated field can determine Profit.

Example formula:

```
[Sales] - [Cost]
```

A **regular field** is generally a field that already exists in the connected data source, such as Sales, Region, or Order Date. A calculated field is derived from a formula, while a regular field comes directly from the source data. Calculated fields can be edited by changing their formulas; regular fields usually reflect the values stored in the source.

2) Explain the difference between Row-Level and Aggregate-Level calculations with examples.

Row-level calculation: The formula is evaluated separately for each row of data. It can create a result for every individual record. For example, to calculate profit for each order:

```
[Sales] - [Cost]
```

If one row has Sales = 500 and Cost = 300, the calculated Profit for that row is 200.

Aggregate-level calculation: The formula uses aggregated values, such as SUM, AVG, MIN, or MAX, to calculate a result at the level of detail shown in the visualization. For example, total profit can be calculated as:

```
SUM([Sales]) - SUM([Cost])
```

If the displayed view contains several orders, Tableau sums Sales and Cost for those orders and subtracts the totals. For instance, total Sales of 1,500 minus total Cost of 900 gives total Profit of 600.

Row-Level	Aggregate-Level
Evaluated for each individual record.	Evaluated using aggregated values in the view.
Example: [Sales] - [Cost]	Example: SUM([Sales]) - SUM([Cost])

3) How does Tableau handle null values in calculated fields? Illustrate with an example.

A **NULL** value means that data is missing, unknown, or unavailable. In Tableau, calculations involving NULL often return NULL, and NULL values may be excluded from some aggregate calculations. The exact result depends on the function and calculation used. To handle NULL values, Tableau provides functions such as **IFNULL()** and **ZN()**.

Example 1 — Replace a NULL value:

```
IFNULL([Discount], 0)
```

If Discount is NULL, this calculation returns 0; otherwise, it returns the existing Discount value.

Example 2 — Treat a NULL numeric value as zero:

```
ZN([Sales])
```

If Sales is NULL, ZN() returns 0. This can be useful when a missing numeric value should be treated as zero. However, replacing NULL with zero should only be done when zero is an appropriate meaning for the missing data.

Conclusion

Calculated fields allow Tableau users to create new information from existing data. Row-level calculations work on individual records, while aggregate-level calculations work on summarized values. NULL-handling functions such as IFNULL() and ZN() help manage missing values and make calculations more reliable.